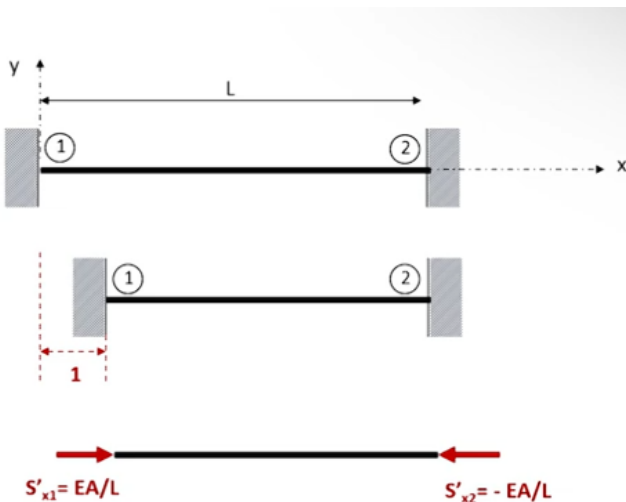
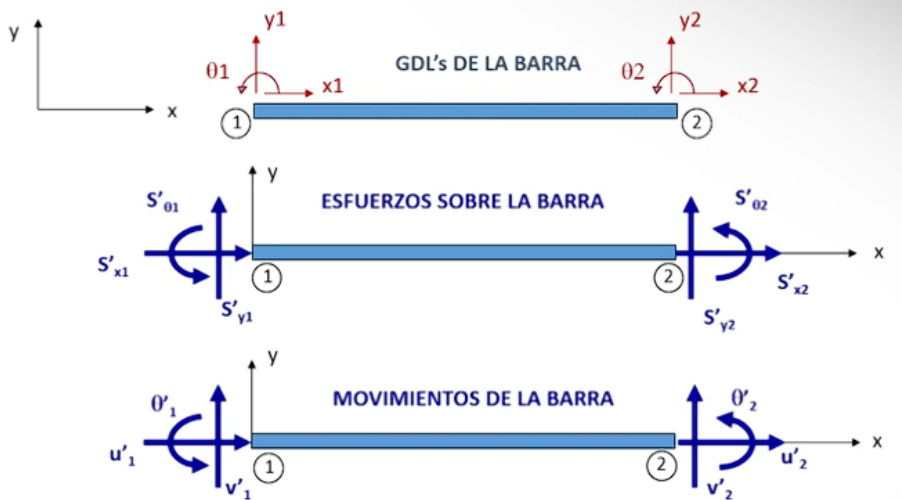


Matricial

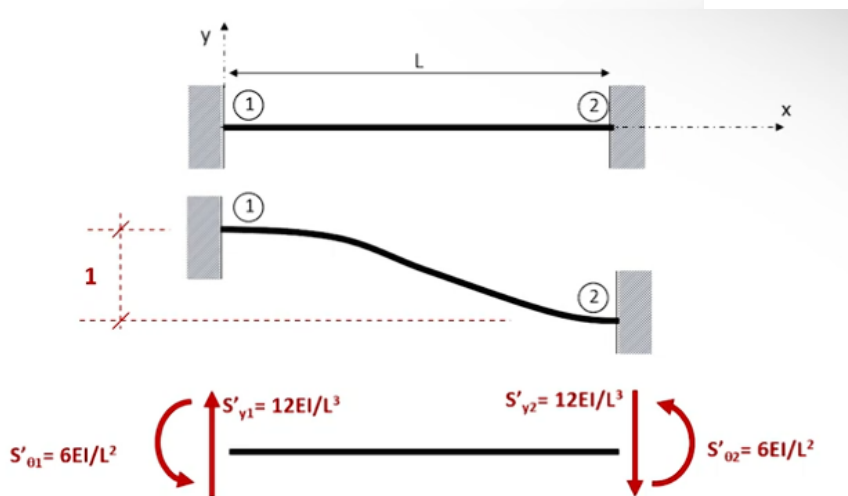
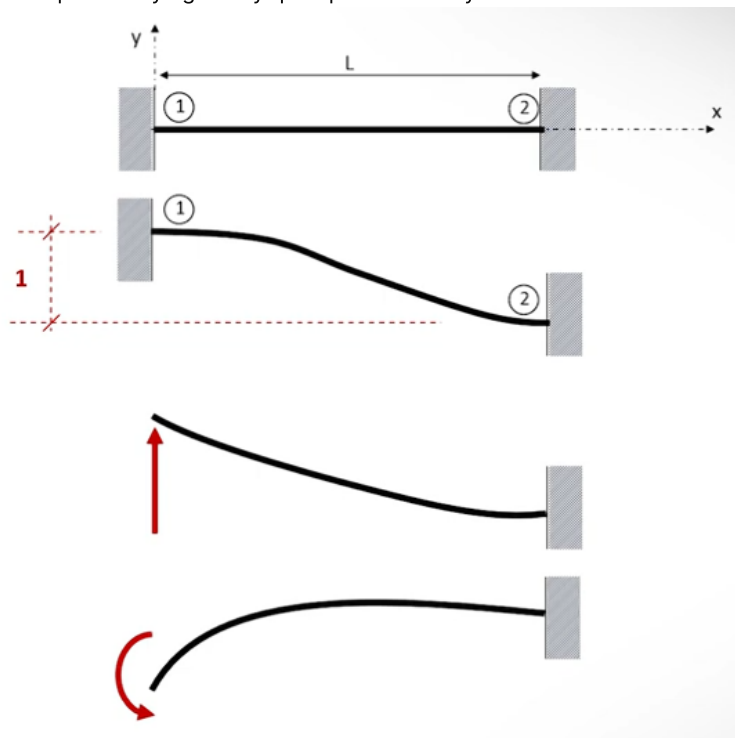
1. Sistema de ejes globales
2. Numeración de nodos
3. Numeración de las barras
4. Sistema de ejes locales en cada barra
5. Selección y numeración de grados de libertad
6. Matriz de rigidez de cada elemento
 - 6.1 en ejes locales
 - 6.2 en ejes globales
7. Ensamblaje de las matrices de rigidez
8. Vector de cargas
9. Condiciones de contorno
10. Resolución del sistema de ecuaciones
11. Cálculo de esfuerzos
12. Cálculo de las reacciones



$$\Delta L = 1 = \frac{N}{EA} L$$

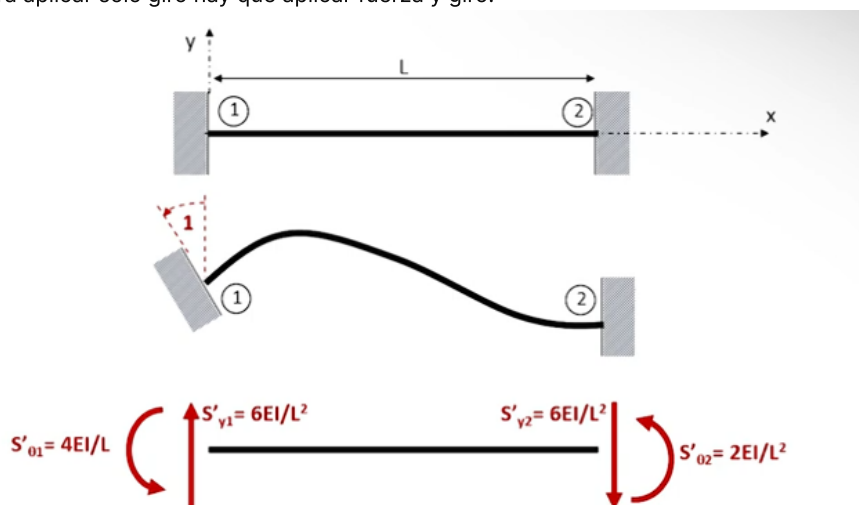
Ec desplazamiento vertical unitario sin giro

Para que no haya giro hay que aplicar fuerza y momento



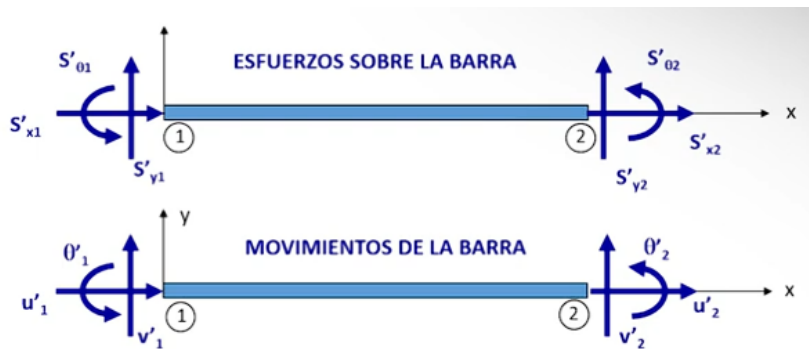
Ec giro unitario sin desplazamiento

Para aplicar solo giro hay que aplicar fuerza y giro:



Ec de equilibrio

Axiles



AXILES

$$\left. \begin{array}{l} \text{Cambio de longitud} \quad L^* = L + (u'_2 - u'_1) \\ \text{Equilibrio según x} \quad S'_{x1} + S'_{x2} = 0 \\ u'_2 - u'_1 = S'_{x2} L / (EA) \end{array} \right\} \begin{array}{l} S'_{x2} = \frac{EA}{L} (u'_2 - u'_1) \\ S'_{x1} = \frac{EA}{L} (u'_1 - u'_2) \end{array}$$

Cortantes y Momentos

MOMENTOS

$$S'_{\theta1} = \frac{4EI}{L} \theta'_1 + \frac{2EI}{L} \theta'_2 + \frac{6EI}{L^2} (v'_1 - v'_2)$$

$$S'_{\theta2} = \frac{2EI}{L} \theta'_1 + \frac{4EI}{L} \theta'_2 + \frac{6EI}{L^2} (v'_1 - v'_2)$$

CORTANTES

$$S'_{y1} = \frac{S'_{\theta1} - S'_{\theta2}}{L} = -S'_{y2}$$

$$S'_{y1} = \frac{6EI}{L^2} \theta'_1 + \frac{6EI}{L^2} \theta'_2 + \frac{12EI}{L^3} (v'_1 - v'_2)$$

$$S'_{y2} = -\frac{6EI}{L^2} \theta'_1 - \frac{6EI}{L^2} \theta'_2 - \frac{12EI}{L^3} (v'_1 - v'_2)$$

Sistema

$$\begin{Bmatrix} S'_{x1} \\ S'_{y1} \\ S'_{\theta1} \end{Bmatrix} = \begin{bmatrix} \frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \begin{Bmatrix} u'_1 \\ v'_1 \\ \theta'_1 \end{Bmatrix} + \begin{bmatrix} -\frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \end{bmatrix} \begin{Bmatrix} u'_2 \\ v'_2 \\ \theta'_2 \end{Bmatrix}$$

$$\begin{Bmatrix} S'_{x2} \\ S'_{y2} \\ S'_{\theta2} \end{Bmatrix} = \begin{bmatrix} -\frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} \end{bmatrix} \begin{Bmatrix} u'_1 \\ v'_1 \\ \theta'_1 \end{Bmatrix} + \begin{bmatrix} \frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \begin{Bmatrix} u'_2 \\ v'_2 \\ \theta'_2 \end{Bmatrix}$$

$$S'_1 = \begin{Bmatrix} S'_{x1} \\ S'_{y1} \\ S'_{\theta1} \end{Bmatrix} \quad S'_2 = \begin{Bmatrix} S'_{x2} \\ S'_{y2} \\ S'_{\theta2} \end{Bmatrix} \quad d'_1 = \begin{Bmatrix} u'_1 \\ v'_1 \\ \theta'_1 \end{Bmatrix} \quad d'_2 = \begin{Bmatrix} u'_2 \\ v'_2 \\ \theta'_2 \end{Bmatrix}$$

$$\begin{Bmatrix} S'_1 \\ S'_2 \end{Bmatrix} = \begin{bmatrix} K'_{11} & K'_{12} \\ K'_{21} & K'_{22} \end{bmatrix} \begin{Bmatrix} d'_1 \\ d'_2 \end{Bmatrix}$$

$$\{S'\} = \begin{Bmatrix} S'_1 \\ S'_2 \end{Bmatrix}$$

$$\{d'\} = \begin{Bmatrix} d'_1 \\ d'_2 \end{Bmatrix}$$

$$[K'] = \begin{bmatrix} K'_{11} & K'_{12} \\ K'_{21} & K'_{22} \end{bmatrix}$$

$$\{S'\} = [K'] \{d'\}$$

$$\{S'\} = [K']\{d'\}$$

$$\begin{aligned} \{S\} &= [T]\{S'\} \\ \{d\} &= [T]\{d'\} \end{aligned} \Rightarrow [T]^{-1} = [T]^T \Rightarrow \begin{aligned} \{S'\} &= [T]^T \{S\} \\ \{d'\} &= [T]^T \{d\} \end{aligned}$$

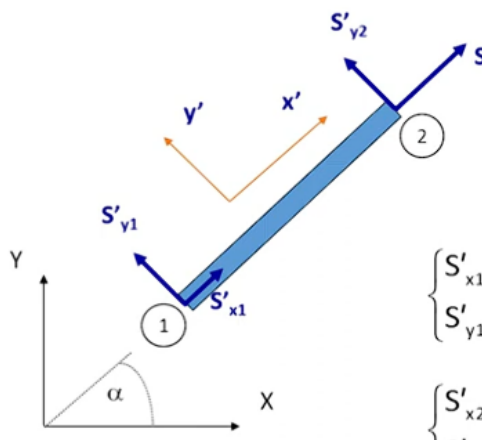
$$\begin{aligned} \{S'_1\} &= [K'_{11}]\{d'_1\} + [K'_{12}]\{d'_2\} \\ \{S'_2\} &= [K'_{21}]\{d'_1\} + [K'_{22}]\{d'_2\} \end{aligned} \Rightarrow \begin{aligned} \{S'_1\} &= [K'_{11}] \cdot [T]^T \{d_1\} + [K'_{12}] \cdot [T]^T \{d_2\} \\ \{S'_2\} &= [K'_{21}] \cdot [T]^T \{d_1\} + [K'_{22}] \cdot [T]^T \{d_2\} \end{aligned}$$

$$\begin{aligned} \{S_1\} &= [T] \cdot \{S'_1\} = [T] \cdot [K'_{11}] \cdot [T]^T \{d_1\} + [T] \cdot [K'_{12}] \cdot [T]^T \{d_2\} \\ \{S_2\} &= [T] \cdot \{S'_2\} = [T] \cdot [K'_{21}] \cdot [T]^T \{d_1\} + [T] \cdot [K'_{22}] \cdot [T]^T \{d_2\} \end{aligned}$$

$$[K_{ij}] = [T] \cdot [K'_{ij}] \cdot [T]^T$$

$$\{S\} = [K]\{d\}$$

Calculo matricial para estructuras articuladas



$$\begin{aligned} \begin{Bmatrix} S'_{x1} \\ S'_{y1} \end{Bmatrix} &= \begin{bmatrix} \frac{EA}{L} & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} u'_1 \\ v'_1 \end{Bmatrix} + \begin{bmatrix} -\frac{EA}{L} & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} u'_2 \\ v'_2 \end{Bmatrix} \\ \begin{Bmatrix} S'_{x2} \\ S'_{y2} \end{Bmatrix} &= \begin{bmatrix} -\frac{EA}{L} & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} u'_1 \\ v'_1 \end{Bmatrix} + \begin{bmatrix} \frac{EA}{L} & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} u'_2 \\ v'_2 \end{Bmatrix} \end{aligned}$$

K'_{11} K'_{12}
 K'_{21} K'_{22}

$$\{S_1\} = [T] \cdot [K'_{11}] \cdot [T]^T \{d_1\} + [T] \cdot [K'_{12}] \cdot [T]^T \{d_2\}$$

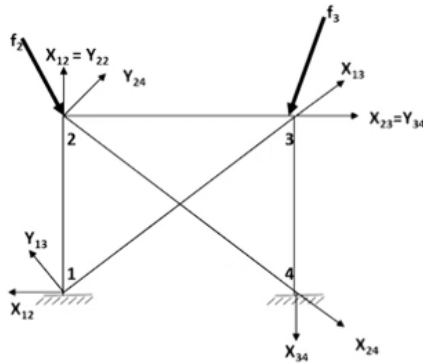
$$\{S_2\} = [T] \cdot [K'_{21}] \cdot [T]^T \{d_1\} + [T] \cdot [K'_{22}] \cdot [T]^T \{d_2\}$$

$$\begin{Bmatrix} S_{X1} \\ S_{Y1} \\ S_{X2} \\ S_{Y2} \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix}$$

$$[K_{11}] = [K_{22}] = \frac{EA}{L} \begin{bmatrix} \cos^2 \alpha & \cos \alpha \sin \alpha \\ \cos \alpha \sin \alpha & \sin^2 \alpha \end{bmatrix}$$

$$[K_{12}] = [K_{21}] = -\frac{EA}{L} \begin{bmatrix} \cos^2 \alpha & \cos \alpha \sin \alpha \\ \cos \alpha \sin \alpha & \sin^2 \alpha \end{bmatrix}$$

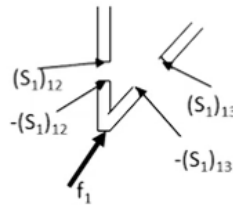
Ensamblar matrices de rigidez en cálculo matricial



Se ha calculado K_{11} , K_{12} , K_{21} y K_{22} para cada barra

$$S_1 = K_{11} d_1 + K_{12} d_2$$

$$S_2 = K_{21} d_1 + K_{22} d_2$$



$$f_1 - (S_1)_{12} - (S_1)_{13} = 0$$

$$f_1 = (S_1)_{12} + (S_1)_{13} = (K_{11})_{12}d_1 + (K_{12})_{12}d_2 + (K_{11})_{13}d_1 + (K_{12})_{13}d_3$$

$$\begin{Bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{Bmatrix} = \begin{bmatrix} (K_{11})_{12} + (K_{11})_{13} & (K_{12})_{12} & (K_{12})_{13} & 0 \\ (K_{22})_{12} + (K_{11})_{23} + (K_{11})_{24} & (K_{12})_{23} & (K_{12})_{24} & (K_{12})_{24} \\ (K_{21})_{13} & (K_{21})_{23} & (K_{22})_{13} + (K_{11})_{24} + (K_{11})_{34} & (K_{12})_{34} \\ 0 & (K_{21})_{24} & (K_{21})_{34} & (K_{22})_{24} + (K_{22})_{34} \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{Bmatrix}$$

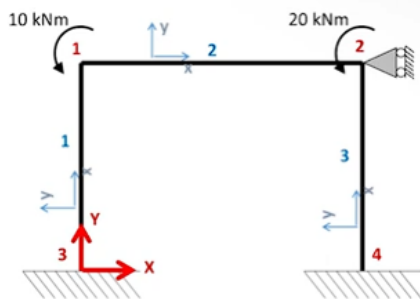
$$f_1 = (K_{11})_{12}d_1 + (K_{12})_{12}d_2 + (K_{11})_{13}d_1 + (K_{12})_{13}d_3$$

$$f_2 = (K_{21})_{12}d_1 + (K_{22})_{12}d_2 + (K_{11})_{23}d_2 + (K_{12})_{23}d_3 + (K_{11})_{24}d_2 + (K_{12})_{24}d_4$$

$$f_3 = (K_{21})_{13}d_1 + (K_{22})_{13}d_3 + (K_{21})_{23}d_2 + (K_{22})_{23}d_3 + (K_{11})_{34}d_3 + (K_{12})_{34}d_4$$

$$f_4 = (K_{21})_{24}d_2 + (K_{22})_{14}d_4 + (K_{21})_{34}d_3 + (K_{22})_{34}d_4$$

MATRIZ SIMÉTRICA



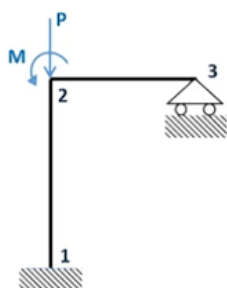
$[K_{11}]_2 + [K_{22}]_1$	$[K_{12}]_2$	$[K_{21}]_1$	0
$[K_{21}]_2$	$[K_{22}]_2 + [K_{22}]_3$	0	$[K_{21}]_3$
$[K_{12}]_1$	0	$[K_{11}]_1$	0
0	$[K_{12}]_3$	0	$[K_{11}]_3$

8. CONSTRUCCIÓN DEL VECTOR DE CARGAS

Sólo cargas en los nodos

Criterios de signos (ejes globales)

Reacciones son cargas de valor desconocido

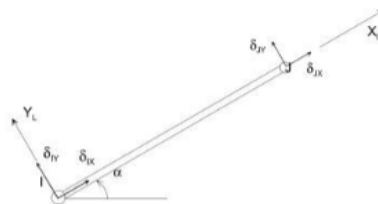


$$\{f_2\} = \begin{Bmatrix} 0 \\ -P \\ M \end{Bmatrix}$$

$$\{f\} = \begin{Bmatrix} f_1 \\ f_2 \\ f_3 \end{Bmatrix} = \begin{Bmatrix} R_{x1} \\ R_{y1} \\ M_1 \\ \hline 0 \\ -P \\ M \\ \hline R_{x3} \\ R_{y3} \\ M_3 \end{Bmatrix}$$

BARRA BIARTICULADA PLANA- RIGIDEZ EN EL SISTEMA LOCAL

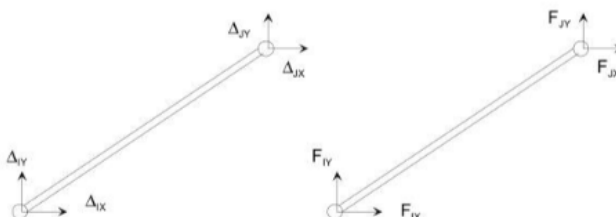
$$\begin{Bmatrix} P_{IX} \\ P_{IY} \\ P_{JX} \\ P_{JY} \end{Bmatrix} = \begin{bmatrix} EA/L & 0 & -EA/L & 0 \\ 0 & 0 & 0 & 0 \\ -EA/L & 0 & EA/L & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \delta_{IX} \\ \delta_{IY} \\ \delta_{JX} \\ \delta_{JY} \end{Bmatrix}$$



BARRA BIARTICULADA PLANA- RIGIDEZ EN EL SISTEMA GENERAL

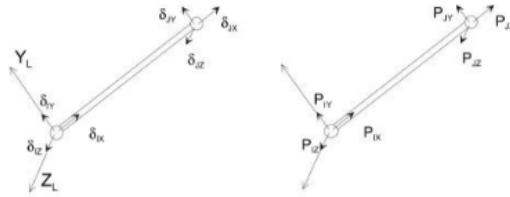
$$\begin{Bmatrix} F_{IX} \\ F_{IY} \\ F_{JX} \\ F_{JY} \end{Bmatrix} = \frac{EA}{L} \begin{bmatrix} c^2 & sc & -c^2 & -sc \\ sc & s^2 & -sc & -s^2 \\ -c^2 & -sc & c^2 & sc \\ -sc & -s^2 & sc & s^2 \end{bmatrix} \begin{Bmatrix} \Delta_{IX} \\ \Delta_{IY} \\ \Delta_{JX} \\ \Delta_{JY} \end{Bmatrix}$$

$$s = \text{seno } \alpha \\ c = \text{coseno } \alpha$$



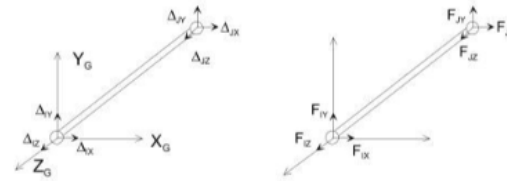
BARRA BIARTICULADA ESPACIAL - RIGIDEZ EN EL SISTEMA LOCAL

$$\begin{Bmatrix} P_{IX} \\ P_{IY} \\ P_{IZ} \\ P_{JX} \\ P_{JY} \\ P_{JZ} \end{Bmatrix} = \frac{EA}{L} \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \delta_{IX} \\ \delta_{IY} \\ \delta_{IZ} \\ \delta_{JX} \\ \delta_{JY} \\ \delta_{JZ} \end{Bmatrix}$$



BARRA BIARTICULADA ESPACIAL - RIGIDEZ EN EL SISTEMA GENERAL

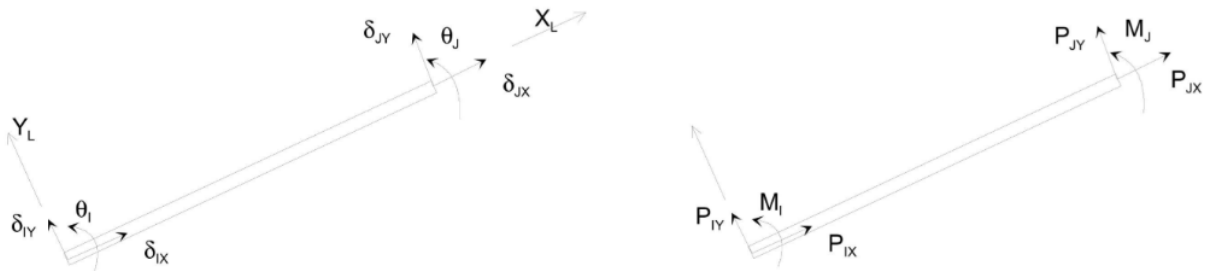
$$\begin{Bmatrix} F_{IX} \\ F_{IY} \\ F_{IZ} \\ F_{JX} \\ F_{JY} \\ F_{JZ} \end{Bmatrix} = EA/L \begin{bmatrix} \lambda\lambda & \lambda\mu & \lambda\nu & -\lambda\lambda & -\lambda\mu & -\lambda\nu \\ \mu\lambda & \mu\mu & \mu\nu & -\mu\lambda & -\mu\mu & -\mu\nu \\ \nu\lambda & \nu\mu & \nu\nu & -\nu\lambda & -\nu\mu & -\nu\nu \\ -\lambda\lambda & -\lambda\mu & -\lambda\nu & \lambda\lambda & \lambda\mu & \lambda\nu \\ -\mu\lambda & -\mu\mu & -\mu\nu & \mu\lambda & \mu\mu & \mu\nu \\ -\nu\lambda & -\nu\mu & -\nu\nu & \nu\lambda & \nu\mu & \nu\nu \end{bmatrix} \begin{Bmatrix} \Delta_{IX} \\ \Delta_{IY} \\ \Delta_{IZ} \\ \Delta_{JX} \\ \Delta_{JY} \\ \Delta_{JZ} \end{Bmatrix}$$



λ = coseno director eje X_L
 μ = coseno director eje Y_L
 ν = coseno director eje Z_L

VIGA A FLEXIÓN EN EL PLANO - RIGIDEZ EN EL SISTEMA LOCAL

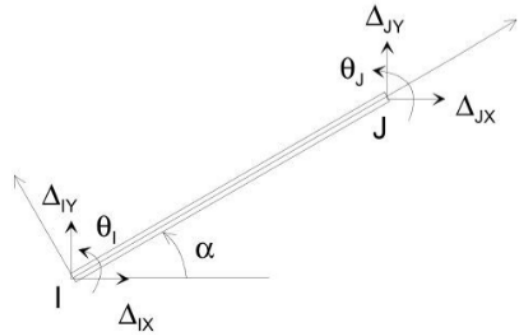
$$\begin{Bmatrix} P_{IX} \\ P_{IY} \\ M_I \\ P_{JX} \\ P_{JY} \\ M_J \end{Bmatrix} = \begin{bmatrix} EA/L & 0 & 0 & -EA/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -EA/L & 0 & 0 & EA/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix} \begin{Bmatrix} \delta_{IX} \\ \delta_{IY} \\ \theta_I \\ \delta_{JX} \\ \delta_{JY} \\ \theta_J \end{Bmatrix}$$



VIGA A FLEXIÓN EN EL PLANO - RIGIDEZ EN EL SISTEMA GENERAL

$$\begin{Bmatrix} F_{IX} \\ F_{IY} \\ M_I \\ F_{JX} \\ F_{JY} \\ M_J \end{Bmatrix} = \begin{bmatrix} EAc^2/L & EAsc/L & -6EIs/L^2 & -EAc^2/L & -EAsc/L & -6EIs/L^2 \\ +12EIs^2/L^3 & -12EIs/L^3 & & -12EIs^2/L^3 & +12EIs/L^3 & \\ EAsc/L & EAs^2/L & 6EIs/L^2 & -EAsc/L & -EAs^2/L & 6EIs/L^2 \\ -12EIs/L^3 & +12EIs^2/L^3 & & +12EIs/L^3 & -12EIs^2/L^3 & \\ -6EIs/L^2 & 6EIs/L^2 & 4EI/L & 6EIs/L^2 & -6EIs/L^2 & 2EI/L \\ -EAc^2/L & -EAsc/L & 6EIs/L^2 & EAc^2/L & EAsc/L & 6EIs/L^2 \\ -12EIs^2/L^3 & +12EIs/L^3 & & +12EIs^2/L^3 & -12EIs/L^3 & \\ -EAsc/L & -EAs^2/L & -6EIs/L^2 & EAsc/L & EAs^2/L & -6EIs/L^2 \\ +12EIs/L^3 & -12EIs^2/L^3 & & +12EIs/L^3 & -12EIs^2/L^3 & \\ -6EIs/L^2 & 6EIs/L^2 & 2EI/L & 6EIs/L^2 & -6EIs/L^2 & 4EI/L \end{bmatrix} \begin{Bmatrix} \Delta_{IX} \\ \Delta_{IY} \\ \theta_I \\ \Delta_{JX} \\ \Delta_{JY} \\ \theta_J \end{Bmatrix}$$

$$c = \cos \alpha \quad s = \sin \alpha$$



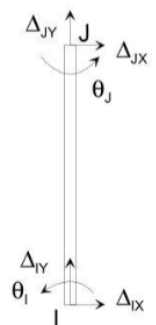
VIGA A FLEXION EN EL PLANO. VIGA HORIZONTAL

$$\begin{Bmatrix} F_{IX} \\ F_{IY} \\ M_I \\ F_{JX} \\ F_{JY} \\ M_J \end{Bmatrix} = \begin{bmatrix} EA/L & 0 & 0 & -EA/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -EA/L & 0 & 0 & EA/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix} \begin{Bmatrix} \Delta_{IX} \\ \Delta_{IY} \\ \theta_I \\ \Delta_{JX} \\ \Delta_{JY} \\ \theta_J \end{Bmatrix}$$



VIGA A FLEXION EN EL PLANO. VIGA VERTICAL

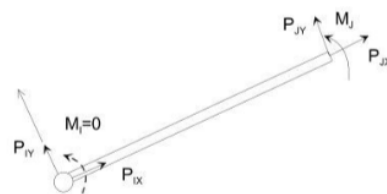
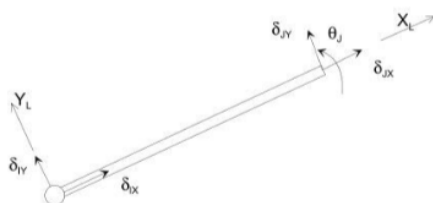
$$\begin{Bmatrix} F_{IX} \\ F_{IY} \\ M_I \\ F_{JX} \\ F_{JY} \\ M_J \end{Bmatrix} = \begin{bmatrix} 12EI/L^3 & 0 & -6EI/L^2 & -12EI/L^3 & 0 & -6EI/L^2 \\ 0 & EA/L & 0 & 0 & -EA/L & 0 \\ -6EI/L^2 & 0 & 4EI/L & 6EI/L^2 & 0 & 2EI/L \\ -12EI/L^3 & 0 & 6EI/L^2 & 12EI/L^3 & 0 & 6EI/L^2 \\ 0 & -EA/L & 0 & 0 & EA/L & 0 \\ -6EI/L^2 & 0 & 2EI/L & 6EI/L^2 & 0 & 4EI/L \end{bmatrix} \begin{Bmatrix} \Delta_{IX} \\ \Delta_{IY} \\ \theta_I \\ \Delta_{JX} \\ \Delta_{JY} \\ \theta_J \end{Bmatrix}$$



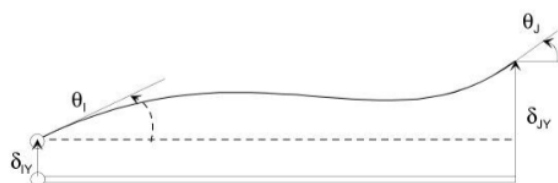
Casos

ELEMENTO ARTICULADO - EMPOTRADO - RIGIDEZ EN EL SISTEMA LOCAL

$$\begin{Bmatrix} P_{IX} \\ P_{IY} \\ M_I \\ P_{JX} \\ P_{JY} \\ M_J \end{Bmatrix} = \begin{bmatrix} EA/L & 0 & 0 & -EA/L & 0 & 0 \\ 0 & 3EI/L^3 & 0 & 0 & -3EI/L^3 & 3EI/L^2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -EA/L & 0 & 0 & EA/L & 0 & 0 \\ 0 & -3EI/L^3 & 0 & 0 & 3EI/L^3 & -3EI/L^2 \\ 0 & 3EI/L^2 & 0 & 0 & -3EI/L^2 & 3EI/L \end{bmatrix} \begin{Bmatrix} \delta_{IX} \\ \delta_{IY} \\ \theta_I \\ \delta_{JX} \\ \delta_{JY} \\ \theta_J \end{Bmatrix}$$



Giro en la articulación $\theta_I = \frac{3}{2L}(\delta_{JY} - \delta_{IY}) - \frac{\theta_J}{2}$



ELEMENTO ARTICULADO - EMPOTRADO - RIGIDEZ EN EL SISTEMA GENERAL

$$\begin{Bmatrix} F_{IX} \\ F_{IY} \\ M_I \\ F_{JX} \\ F_{JY} \\ M_J \end{Bmatrix} = \begin{bmatrix} EAc^2/L & EAsc/L & 0 & -EAc^2/L & -EAsc/L & -3EIs/L^2 \\ +3EIs^2/L^3 & -3EIs^2/L^3 & 0 & -3EIs^2/L^3 & +3EIs^2/L^3 & 3EIs/L^2 \\ EAsc/L & EAs^2/L & 0 & -EAsc/L & -EAs^2/L & 3EIs/L^2 \\ -3EIs^2/L^3 & +3EIs^2/L^3 & 0 & +3EIs^2/L^3 & -3EIs^2/L^3 & 3EIs/L^2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -EAc^2/L & -EAsc/L & 0 & EAc^2/L & EAsc/L & 3EIs/L^2 \\ -3EIs^2/L^3 & +3EIs^2/L^3 & 0 & +3EIs^2/L^3 & -3EIs^2/L^3 & 3EIs/L^2 \\ -EAsc/L & -EAs^2/L & 0 & EAsc/L & EAs^2/L & -3EIs/L^2 \\ +3EIs^2/L^3 & -3EIs^2/L^3 & 0 & -3EIs^2/L^3 & +3EIs^2/L^3 & -3EIs/L^2 \\ -3EIs/L^2 & 3EIs/L^2 & 0 & 3EIs/L^2 & -3EIs/L^2 & 3EI/L \end{bmatrix} \begin{Bmatrix} \Delta_{IX} \\ \Delta_{IY} \\ \theta_I \\ \Delta_{JX} \\ \Delta_{JY} \\ \theta_J \end{Bmatrix}$$

